

CAPSTONE PROJECT

CYBERSHIELD AI MACHINE LEARNING-BASED NETWORK INTRUSION DETECTION SYSTEM

Presented By:

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OUTLINE

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- **System Development Approach**
- **Algorithm & Deployment**
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PROBLEM STATEMENT

- Modern computer networks are continuously exposed to cyberattacks and unauthorized access attempts. Manually monitoring large volumes of network traffic is difficult, time-consuming, and prone to errors.
- Traditional security systems may fail to identify new or complex intrusion patterns efficiently. Therefore, an intelligent system is required to analyze network traffic data and distinguish normal network activity from malicious or anomalous activity.
- The objective is to develop a machine learning-based Network Intrusion Detection System capable of identifying suspicious traffic and supporting early detection of cyber threats.

PROPOSED SOLUTION

- **CyberShield AI – Network Intrusion Detection System**

- The proposed solution uses machine learning to classify network traffic as **Normal** or **Anomaly**.

- **Main components**

- **Dataset Collection:**

Use a Network Intrusion Detection dataset containing normal and malicious network traffic records.

- **Data Upload:**

Upload the training dataset to IBM watsonx.ai as a data asset.

- **Data Preprocessing:**

Process numerical and categorical features such as protocol type, service, flag, source bytes, destination bytes, and connection statistics.

- **Model Training:**

Use IBM watsonx.ai AutoAI to generate and compare multiple machine learning pipelines.

- **Best Model Selection:**

Select the pipeline with the highest cross-validation performance.

- **Model Deployment:**

Deploy the trained model as an online deployment using IBM watsonx.ai Runtime.

- **Frontend Integration:**

Connect a Streamlit application to the IBM deployment API.

- **Prediction:**

Allow users to upload test CSV data and receive Normal or Anomaly predictions.

SYSTEM APPROACH

- **System Requirements**
- **Hardware Requirements**
- Computer or laptop
- Stable internet connection
- Minimum 4 GB RAM
- **Software and Services**
- Python
- IBM Cloud Lite
- IBM watsonx.ai
- IBM watsonx.ai AutoAI
- IBM watsonx.ai Runtime
- Visual Studio Code
- Streamlit
- **Python Libraries**
- Streamlit
- Pandas
- Requests
- python-dotenv
- **System Workflow**
- Dataset → IBM watsonx.ai AutoAI → Model Training → Best Pipeline → Online Deployment → REST API → Streamlit Frontend → Prediction Output

ALGORITHM & DEPLOYMENT

- **Algorithm Selection**
- IBM watsonx.ai AutoAI generated eight machine learning pipelines and compared them using cross-validation performance.
- The best-performing model was:
- **Algorithm:** Snap Decision Tree Classifier
- **Prediction Type:** Binary Classification
- **Target Column:** class
- **Classes:** Normal and Anomaly
- **Accuracy:** 0.995 or 99.5%
- **Data Input**
- The model uses network traffic features such as:
- Duration, protocol_type, service, flag, src bytes , dst bytes, connection and host-related statistics
- **Training Process**
- Uploaded Train_data.csv to IBM watsonx.ai
- Selected class as the prediction column
- Disabled time-series analysis
- Ran the AutoAI experiment
- Compared eight generated pipelines
- Selected the highest-ranked pipeline
- **Deployment Process**
- Saved the best pipeline as a model
- Created a deployment space
- Created an online deployment
- Generated a secured public REST API endpoint
- Connected the endpoint to the Streamlit frontend
-

RESULT

- The IBM watsonx.ai AutoAI experiment successfully generated eight machine learning pipelines.
- **Final Results**
- Best Model: Snap Decision Tree Classifier
- Cross-Validation Accuracy: 0.995
- Output Classes: Normal and Anomaly
- Deployment Status: Online and Deployed
- Frontend: Streamlit application
- Input Method: CSV file upload
- Prediction Output: Classification and probability values
- The deployed model successfully classified test network traffic records through the Streamlit interface.

SCREENSHOTS REGARDING DEPLOYMENTS

IBM watsonx

Upgrade

🔔

Anwesha Sural's Account

Sydney

AS

Projects / CyberShield AI: ML-Oriented Network Intrusion ...

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Add users as collaborators

→

📊

Add data to work with

→

💬

Chat and build prompts with foundation models

→

🔧

Tune a foundation model with labeled data

→

View all

Collapse

Jump back in

By all

🔍

P1 - Snap Decision Tree Classifier: CyberShi...

14 hours ago by you

🔍

CyberShield AI - Network Intrusion Detection

14 hours ago by you

🔍

CyberShield AI - Network Intrusion Detection

22 hours ago by you

🔍

CyberShield AI - Network Intrusion Detection

22 hours ago by you

🔍

CyberShield AI - Network Intrusion Detection

2 days ago by you

Resource usage

🔍

For this month in this project

2.2 CUH

8377 Tokens

0 Hosting hours

0 Pages

Your documentation

Get started with your documentation

You can create and manage documents about work that you do in this project.

Open Documentation editor →

Project history

🔄

📌 Asset promoted to space

You promoted P1 - Snap Decision Tree Classifier: CyberShield AI - Network Intrusion Detection from project CyberShield AI: ML-Oriented Network Intrusion Detection System to space CyberShield AI ...

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[Deployment spaces](#) / [CyberShield AI Deployment Space](#) / [P1 - Snap Decision Tree Classifier: CyberShield ...](#) /

CyberShield AI Online Deployment ✔ Deployed Online

API reference

Test

Endpoints for inferencing ^①

Private endpoint

☒ Show serving name
 ☐ Show deployment ID

Bearer <token> ^①

```
https://private.au-syd.ml.cloud.ibm.com/ml/v4/deployments/cybershield_ai/predictions?version=2021-05-01
```

📄

IAM

Public endpoint

```
https://au-syd.ml.cloud.ibm.com/ml/v4/deployments/cybershield_ai/predictions?version=2021-05-01
```

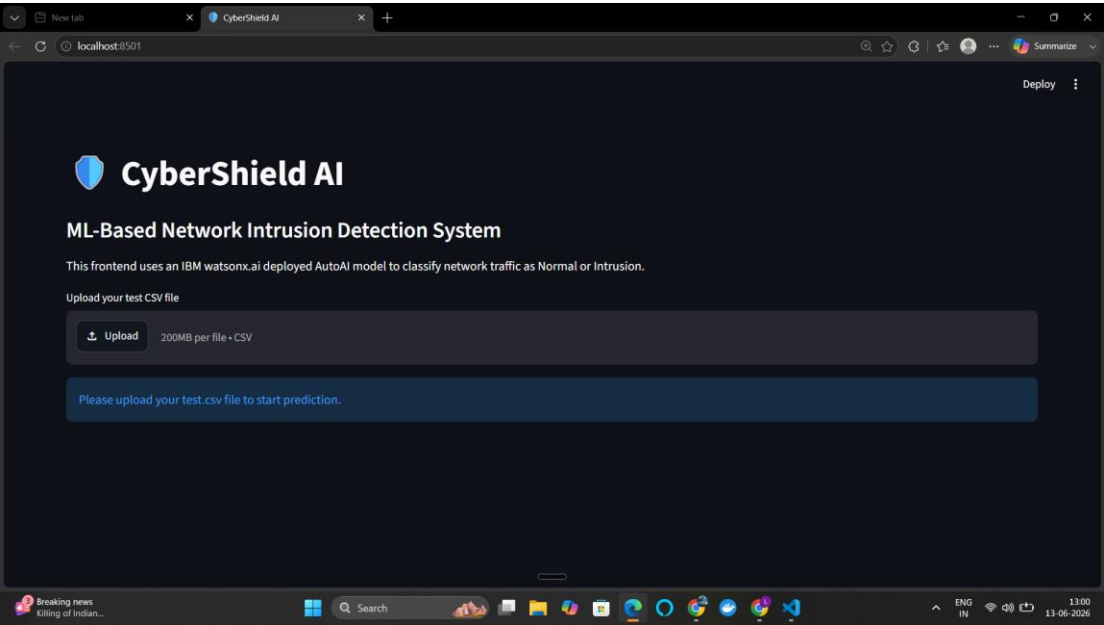
📄

[Learn more](#) about the 2021-05-01 version query parameter

Code snippets

cURL	Java	JavaScript	Python	Scala
# NOTE: you must set \$API_KEY below using information retrieved from your IBM Cloud account (https://au-syd.dai.cloud.ibm.com/docs/content/wsj/analyze-data/ml-authentic) <pre>export API_KEY=<your API key></pre> <pre>export IAM_TOKEN=\$(curl --insecure -X POST --location "https://iam.cloud.ibm.com/identity/token" --header "Content-Type: application/x-www-form-urlencoded" --header "Authorization: Bearer <token>" --data '{"apikey": "<API key>"}</pre>				

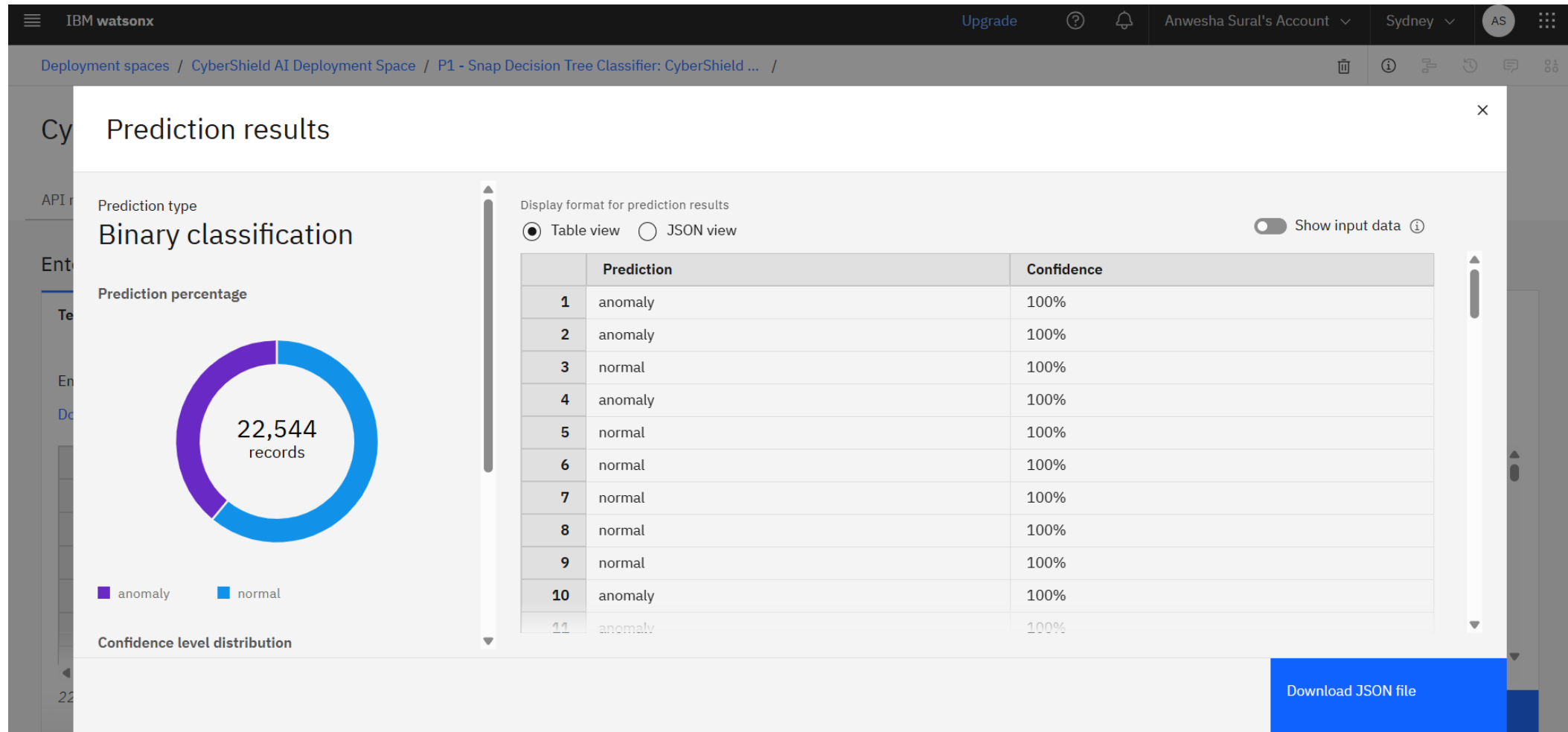
STREAMLIT INTERFACE



Simplified Prediction Output

	0	1
22534	anomaly	1 0
22535	normal	0 1
22536	anomaly	1 0
22537	normal	0 1
22538	anomaly	1 0
22539	normal	0 1
22540	normal	0 1
22541	anomaly	1 0
22542	normal	0 1
22543	normal	0 1

PREDICTION RESULT



CONCLUSION

- CyberShield AI successfully demonstrates an end-to-end machine learning solution for network intrusion detection.
- The system uses IBM watsonx.ai AutoAI to train and evaluate multiple classification pipelines. The best-performing Snap Decision Tree Classifier achieved 99.5% cross-validation accuracy.
- The model was deployed using IBM watsonx.ai Runtime and connected to a Streamlit frontend through a secured API endpoint. Users can upload network traffic data and receive Normal or Anomaly predictions.
- The project demonstrates practical knowledge of machine learning, cybersecurity, cloud deployment, API integration, and frontend development.

FUTURE SCOPE

- **Real-Time Traffic Monitoring:**
Capture and analyze live network packets instead of relying only on uploaded CSV files.
- **Multi-Class Attack Detection:**
Classify attacks into categories such as DoS, Probe, R2L, and U2R.
- **Alert System:**
Send email, SMS, or dashboard notifications when malicious activity is detected.
- **Advanced Models:**
Test Random Forest, XGBoost, neural networks, LSTM, and hybrid deep-learning models.
- **Security Dashboard:**
Display attack counts, risk levels, prediction history, and network-security trends.
- **Continuous Model Retraining:**
Retrain the model using newly collected network traffic data.

REFERENCES

- IBM watsonx.ai Documentation
- IBM watsonx.ai AutoAI Documentation
- IBM Cloud Documentation
- Streamlit Documentation
- Python Documentation
- Pandas Documentation
- Kaggle Network Intrusion Detection Dataset
- AICTE-IBM SkillsBuild

DEPLOYMENT LINK / GITHUB LINK

- Public GitHub Repository
- <https://github.com/lucia-builds/cybershield-ai-network-intrusion-detection.git>
- IBM Cloud Model Deployment Endpoint
https://au-syd.ml.cloud.ibm.com/ml/v4/deployments/cybershield_ai/predictions?version=2021-05-01
- Repository Files
- app.py
- app.json
- requirements.txt
- README.md
- .gitignore
- problemstatement.pdf
- projectpresentation.pptx



THANK YOU